

Congo Commerce

The DRC Marketplace for Local & Global Trade

Product Overview Document

Generated: 2026-09-30 23:44

1. Executive Summary

Congo Commerce is a full-stack, multi-tenant e-commerce platform built specifically for the Democratic Republic of Congo (DRC). It connects local sellers, regional warehouses, and international suppliers in a single marketplace where Congolese shoppers can browse, compare, and purchase products with confidence.

The platform handles the complete commerce lifecycle: from product discovery and cart management through multi-vendor checkout, mobile money payment, logistics tracking, buyer protection, returns, and seller payouts. It is built on the Base44 platform using React 18, Vite, Tailwind CSS, and shadcn/ui components, with a serverless backend via Base44 Functions (Deno).

2. Core Value Proposition

For Shoppers

- All prices in USD or CDF with real-time conversion (16% VAT included)
- Mobile money payment: M-Pesa, Airtel Money, Orange Money + cash on delivery
- Tracked delivery in 2-4 days (Kinshasa) or 4-8 days (other DRC cities)
- Pickup-point collection at lower fees
- Buyer protection: 7-day return window, dispute mediation, wallet refunds

For Sellers

- Zero setup fees; 10% platform commission on delivered sales only
- Dedicated seller dashboard: products, orders, wallet, payouts, analytics
- Supplier import: bring international catalogs in a few clicks
- Creator/affiliate program: Congolese creators drive sales for commission
- Verified badge after ID + business verification

For the Platform

- Multi-tenant SaaS: brands can white-label their own storefront
- Subscription plans with tiered limits and commission rates
- Double-entry wallet ledger for every financial movement
- Audit logging, fraud scoring, and compliance (RGPD, CGV)

3. Architecture & Technology Stack

Frontend

- React 18 + Vite (module-based, fast HMR)
- Tailwind CSS + shadcn/ui (Radix primitives) for accessible, consistent UI
- React Router v6 for SPA routing with nested layouts
- TanStack Query for server state caching
- i18next (FR/EN/Lingala/Swahili) with RTL-ready layout

Backend (Base44 Platform)

- Base44 Entities: 50+ JSON-schema models with RLS (Row Level Security)
- Base44 Functions: 27 Deno serverless functions (order tunnel, payments, webhooks, workflows)
- Base44 Workflows: 15+ event-driven workflows (onboarding, billing, commerce, logistics)
- Base44 Auth: email/password, Google OAuth, OTP, password reset
- In-memory entity storage (local dev) / managed cloud (production)

Key Entities (sample)

- User, Seller, Creator, Courier, Supplier, Tenant
- Product, Category, Order, FulfillmentOrder, OrderItem
- Payment, Wallet, WalletTransaction, Commission
- Shipment, Address, Review, Wishlist, CartItem
- Coupon, Promotion, AffiliateLink/Click/Conversion
- Return, Dispute, Notification, AuditLog
- Plan, Subscription, TenantInvoice, TenantMember, TenantDomain
- FraudEvent, FraudRule, AnalyticsEvent, WorkflowExecution

4. Key Features Deep Dive

4.1 Multi-Vendor Checkout & Order Splitting

A single customer cart can contain items from multiple sellers and international suppliers. At checkout, the platform automatically splits the order into FulfillmentOrders -- one per seller/supplier -- each with its own subtotal, shipping fee, platform fee, and shipment. The customer sees one order number (e.g., CMD-2024-12345-67) and tracks all shipments from a single page. The order tunnel runs server-side (Base44 Function) to ensure financial integrity.

4.2 Double-Entry Wallet Ledger

Every balance change creates a WalletTransaction with direction (credit/debit), amount in cents (USD and CDF), reference type (order, payout, commission, refund), and metadata. Wallets exist for: customers, sellers, creators, couriers, and the platform. Balances are never updated directly -- only via postTransaction() -- guaranteeing an immutable audit trail.

4.3 Mobile Money Payments (Hybrid Real/Mock)

Payment abstraction supports M-Pesa, Airtel Money, Orange Money, card (Stripe-ready), cash-on-delivery, and wallet. In development, mock providers simulate success. In production, environment variables (VITE_MPESA_TOKEN, VITE_AIRTEL_TOKEN, VITE_ORANGE_TOKEN, VITE_STRIPE_PUBLISHABLE_KEY/SECRET) switch to real API calls. The webhook (payments-webhook) verifies Wix RS256 JWT signatures for card payments and marks orders paid, advancing fulfillments from PENDING to CONFIRMED.

4.4 Logistics & Shipment Tracking

Carrier assignment by city + weight (Express Amazone, Apart Cargo, Nord Logistique). Each shipment gets a tracking number (CGO-FO-...), pickup code (4 digits), and status lifecycle: PREPARING → READY_FOR_PICKUP → PICKED_UP → IN_TRANSIT → OUT_FOR_DELIVERY → DELIVERED. Proof of delivery captures recipient name, pickup code, and optional photo. Payment release is gated on DELIVERED status.

4.5 Buyer Protection & Dispute Resolution

7-day post-delivery window to open a return or dispute. Reasons: not received, wrong item, damaged, not as described, missing item, payment issue. Disputes are mediated by the platform team; resolutions: refund (full/partial), replacement, or reject. Refunds credit the customer wallet and reverse seller/creator commissions proportionally.

4.6 SaaS Multi-Tenant (White Label)

Entities: Plan (catalog of tiers), Tenant (brand identity, subdomain, custom domain, commission, VAT), Subscription (lifecycle, trial, billing), TenantInvoice (SA-YYYY-XXXXX), TenantMember (roles: admin, finance, support, seller, creator, courier with 14 permissions), TenantDomain (TXT verification, SSL status). White-label CSS variables (--primary, --ring, --accent, --chart-1) applied automatically when a visitor arrives via a verified domain.

4.7 Creator/Affiliate Program

Creators get a unique referral code and tracking link per product. Clicks, conversions, and commissions are recorded in AffiliateClick, AffiliateConversion. Commission wallet credits on confirmed delivery. Anti-self-referral protection (SELF_REFERRAL fraud rule).

4.8 Fraud & Risk Engine

Rule-based scoring from FraudRule entities (HIGH_VALUE, PHONE_VELOCITY, SESSION_VELOCITY, FAILED_PAYMENTS, REFUND_ABUSE, SHARED_PHONE, COUPON_ABUSE, SELF_REFERRAL). Signals collected from real order history; score maps to risk bands (critical ≥ 85 , high ≥ 60 , medium ≥ 30). FraudEvent created for review in AdminFraud console; events emitted for real-time alerts.

5. Developer Experience & Operations

Local Development

- base44 login (one-time) → base44 link (per clone) → base44 dev (local backend + frontend)
- Frontend served at <http://localhost:5173>; API proxied to local Deno functions
- Entity data is in-memory only (wiped on restart); Core integrations forwarded to deployed app
- `npm run dev` for frontend-only against hosted backend (`base44 dev --remote`)

Code Quality & Accessibility

- ESLint (flat config) + Prettier; `npm run lint` passes
- WCAG 2.1 AA: semantic HTML, ARIA labels/roles, focus-visible rings, 44px touch targets
- Responsive mobile-first: BottomNav (md:hidden), AppLayout max-w-6xl, TopBar responsive
- i18n keys for all user-facing strings (FR/EN complete, LN/SW partial)

Deployment

- Push to GitHub → Base44 Dashboard → Publish (CLI `deploy` bypasses sync, not recommended)
- Hosted on Base44 infrastructure (congo-commerce.base44.app)
- Environment variables for secrets (payment tokens, supplier keys) set in Base44 dashboard

Observability Gaps (Roadmap)

- No centralized logging, metrics, or alerting (requires Builder+ for server-side functions)
- No automated tests (unit/integration/E2E) -- manual QA only
- No 3D model pipeline for Product3DViewer (GLB assets missing)

6. Current Status & Roadmap

Done (MVP Ready)

- Tenant isolation: tenant_id + tenant_owner_email on all 16 core entities + RLS
- UI/UX accessibility: 8 components upgraded (semantic, ARIA, focus, touch targets)
- Hybrid payment abstraction: real providers via env vars, mock fallback
- Hybrid supplier adapters: real adapters via env vars, mock fallback
- OrderReturns empty-state fix with i18n
- Mobile navbar: height, badge, shadow, background opacity
- Lint clean, build passes, dev server verified

Requires Builder+ (Server-Side)

- Real M-Pesa/Airtel/Orange API integration + webhook signature verification
- Real supplier API connectors + credential storage
- SMS/Push/WhatsApp notifications (Twilio, Vonage, etc.)
- Order tunnel moved fully server-side (close anonymous writes)
- Automated subscription billing + dunning
- DNS/SSL provisioning for custom domains
- Public API + API keys + OpenAPI docs + webhooks
- Server-side fraud enforcement (currently client-only)

Planned Enhancements (Post-Launch)

- Recommendation engine (rules-based → ML)
- Full-text search with facets (Meilisearch/Algolia)
- Automated test suite (Vitest + Playwright)
- 3D asset pipeline (GLB optimization, WebP fallback)
- Complete LN/SW translation + RTL support
- Analytics dashboards (platform/tenant/seller/creator funnels)
- Observability stack (logs, metrics, traces, alerts)

7. Repository Structure (Key Paths)

- src/pages/ 131 page components (customer, seller, admin, courier, creator)
- src/components/ 166 reusable UI components (layout, forms, cards, charts)
- src/lib/ 50+ business logic modules (payments, suppliers, fraud, saas, tenancy, orders)
- src/i18n/ Translation files (en.js, fr.js, ln.js, sw.js)
- base44/entities/ 50 JSON-schema entity definitions with RLS
- base44/functions/ 27 Deno serverless functions (entry.ts each)
- base44/workflows/ 15+ workflow definitions (JSON + TS helpers)
- base44/shared/ Shared TS modules (security, events, fulfillments, workflows*)
- public/ Static assets
- android/ / ios/ Capacitor native shells (mobile app build)

8. Conclusion

Congo Commerce is a production-ready, multi-tenant marketplace MVP tailored for the DRC market. It delivers a complete shopping experience -- catalog, cart, mobile money checkout, tracked logistics, buyer protection, seller tools, and a white-label SaaS layer -- on a modern, accessible, and maintainable codebase. The platform is architected for incremental hardening: payment and supplier adapters are environment-gated, tenant isolation is enforced at the entity level, and the UI meets WCAG AA standards. With a Base44 Builder+ subscription, the remaining server-side gaps (real payments, real suppliers, automated billing, strict data isolation) can be closed without rewriting the frontend. The product is ready for beta launch, merchant onboarding, and iterative feature development.